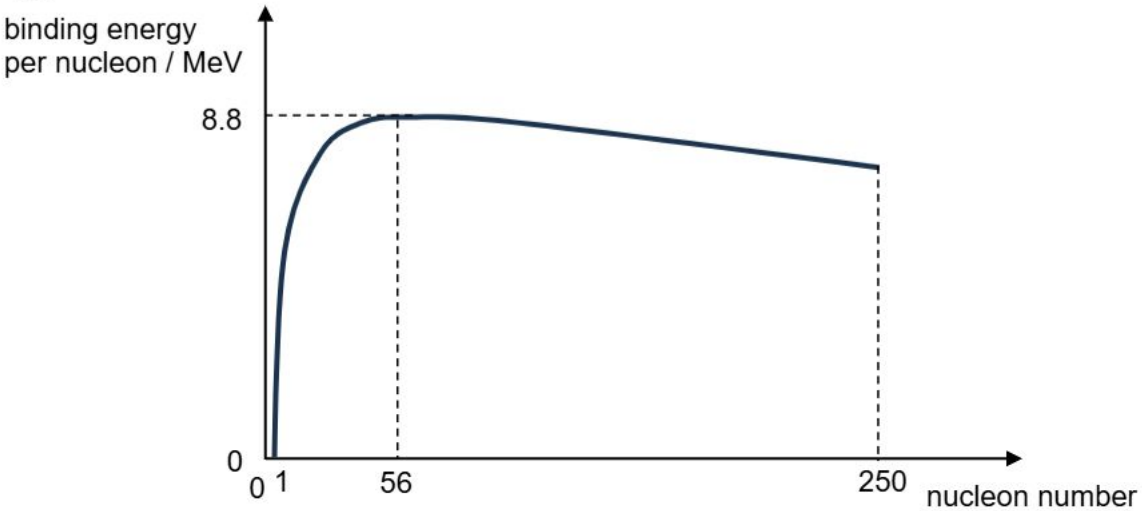
7 **(a)** The binding energy of a nucleus is the <u>energy required to separate to infinity all the nucleons of a nucleus.</u>

(b)



*correct shape of curve, nucleon numbers 1 and 250 and highest point labelled.

(c) The lighter product (daughter) nuclei have higher binding energy per nucleon than the "parent" nucleus.
This means that the formation of the product (daughter) nuclei from their constituent nucleons releases more energy than that required to completely disintegrate the "parent" nucleus.

(d) **(i)** number of α-particles = 0
number of β-particles = 2

(ii) It is not feasible as a tremendous amount of energy is required to overcome the <u>strong electrostatic (Coulomb) repulsion</u> between the nuclei.

(e) decrease in mass = 233.039634 – 136.911557 – 93.915355 – 2(1.008664)
= 0.195394 u

$$\text{energy} = \Delta mc^2$$

$$= 0.195394 \times 1.66 \times 10^{-27} \times \left(3.00 \times 10^{8}\right)^2$$

$$= 2.919186 \times 10^{-11} = 2.92 \times 10^{-11} \text{ J}$$